

ADANI CEMENT · PROCESS OPTIMIZATION

Soft Sensor for Clinker C3S & Free Lime Prediction

A joint engagement to bring near real-time clinker quality prediction and operator guidance to the kiln process, delivered by Faclon Labs on AWS.

Objective

Develop a data-driven Soft Sensor to predict clinker C3S and Free Lime at a frequency of 15–30 minutes using live process data and intra-polated quality data. The solution bridges the delay between clinker production and laboratory analysis, enabling the CCR team to monitor clinker quality in near real time and take corrective actions of process parameters and AFR, before quality deviations become significant.

Deliverables

A. Soft Sensor Model

- Plant-specific soft sensor for prediction of clinker C3S and Free Lime.
- Prediction of C3S and Free Lime every 15–30 minutes.
- Continuous reconciliation of predicted values against laboratory measurements.

Continuous AI-Driven Monitoring

- Real-time display of predicted C3S and Free Lime.
- Predicted vs. actual trend comparison.

Scalable Across Assets and Sites

- Root Cause Analysis for deviations in C3S and Free Lime from their target values.
- Prioritised, operator-friendly recommendations recover clinker quality as per the operator defined priorities within the desired operating limits.

Scope of Work

Historical Data Assessment

- Collect and validate historical process, quality and laboratory data required for model development.
- Review data availability across kiln process parameters and quality control data of kiln feed, clinker, hotmeal chemistry, degree of calcination, fuel mix including fossil fuel and AFR.

Soft Sensor Model Development

- Prepare and clean historical data for model development.
- Develop machine learning models to predict clinker C3S and Free Lime using process and quality parameters.
- Validate model performance using historical laboratory results.
- Continuously reconcile predicted values with laboratory measurements to monitor prediction accuracy and support model improvement.

Online Prediction & Operator Guidance

- Integrate real-time process and quality data (where available) for online prediction.
- Generate predicted C3S and Free Lime values every 15–30 minutes and track predicted and actual values in absolute terms as well as the trend.
- Perform Root Cause Analysis (RCA) guidance for quality deviations based on the corresponding kiln operating conditions.
- Generate operator-friendly recommendations to recover clinker quality as per the operator defined priorities within the desired operating limits.

NOTES & ASSUMPTIONS

- Historical process, fuel, raw meal chemistry and laboratory quality data required for model development will be available.
- The prediction frequency (15 or 30 minutes) will be finalized jointly during implementation based on data availability and business requirements.
- Target values, USL and LSL for C3S and Free Lime will be provided by the plant quality team.
- Recommendations are advisory in nature and intended to support operator decision-making. The model outputs, root cause analysis and recommendations should be jointly reviewed and validated with the plant operations team, Adani process experts and Faclon SMEs before implementation. The plant team will retain responsibility for operational decisions and execution.